

WASP-10b

R-1769

Benchmark run · workflow W8-benchmark-deep · record deep-bench_WASP-10_190826 · created 2026-08-02T21:56:25+00:00

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2458721.6902 +/- 0.0044 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.151 +/- 0.061
Transit depth (Rp/Rs)^2	2.28 +/- 1.84 [%]
Orbital Inclination (inc)	89.14 +/- 1.93
Airmass coefficient 1 (a1)	0.65 +/- 0.44
Airmass coefficient 2 (a2)	0.23 +/- 0.25
Scatter in the residuals of the lightcurve fit is	69.22 %
Best Comparison Star	#3 - [147, 56]
Optimal Aperture	0.87
Optimal Annulus	3.75
Transit Duration (day)	0.087 +/- 0.012

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.151 ± 0.061 vs 0.1592 (deviation 0.13σ)
Duration fit vs published	0.087 d vs 0.0946 d (deviation 0.63σ)
Mid-time O–C	-10.1 min
Depth SNR	2.5
Residual scatter	69.22 %

Light curve

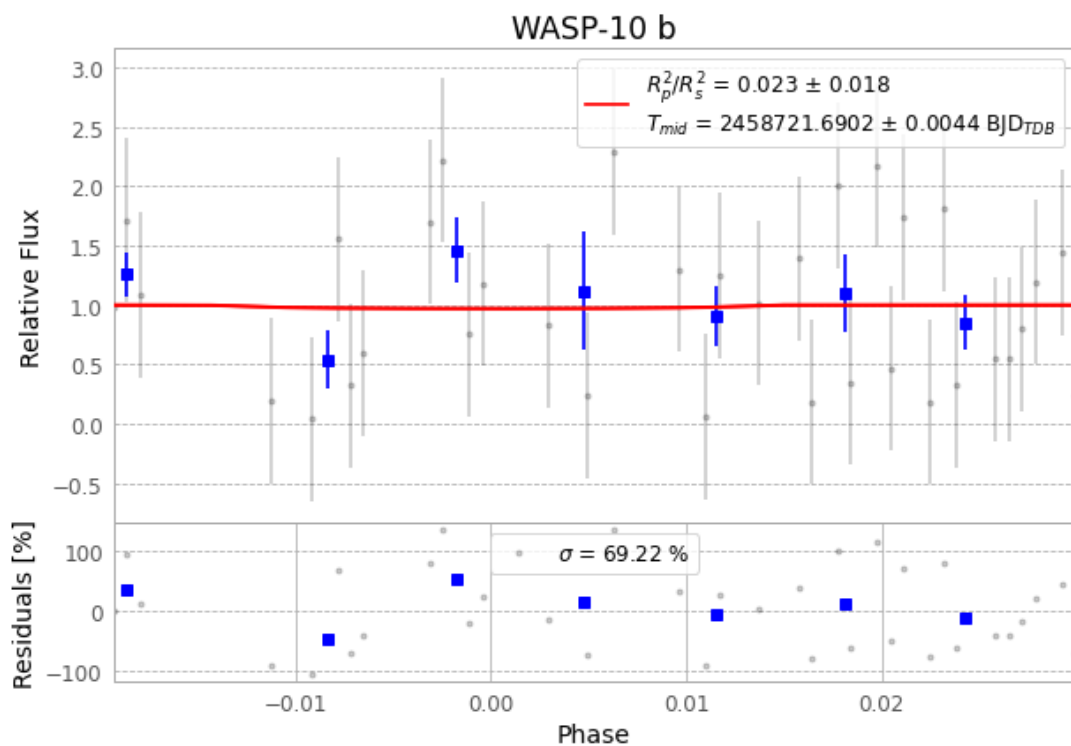


Figure 1 — FinalLightCurve_WASP-10 b_2019-08-25.png (SHA-256 58b6c030c1d416a9...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [471, 134], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 9b76f7482172a52f...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b

Data provenance

76 FITS frames (50 MB), WASP-10190826030012.FITS → WASP-10190826064512.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-10-190826.log (415d81c3c620...), FinalLightCurve_WASP-10 b_2019-08-25.png (58b6c030c1d4...), FinalLightCurve_WASP-10 b_2019-08-25.csv (45236f878152...)

Compute resources

Wall time 155.3 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-02 21:56Z.